Yantong Liu

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EDUCATION

Hong Kong University of Science and Technology (Guangzhou) — **GPA**: 3.78/4.3
MPhil in Computational Media and Arts — *Sep. 2025 - Present*
Advisors: Prof. Mingming Fan and Prof. Chen Liang

Communication University of China — **GPA**: 3.81/4.0
B.A. in Data Journalism — *Sep. 2021 - Jun. 2025*

PUBLICATIONS

Published

- POReviewer: A Visualization System for Retrospective Public Opinion Analysis. Zeyu Li, Manlu Zhou, **Yantong Liu**, Changchang Ye, Xiaomeng Zhou & Jiawan Zhang. *Journal of Visualization*, 2026. DOI: 10.1007/s12650-026-01140-6.
- A Global Election Situation Awareness System. Zhongxin Shan, **Yantong Liu**, Yue Li, Shuqing Wang, Yibin Hou, Zeyu Li. Poster presented at *ChinaVis 2024*, Hong Kong.

Ongoing (*Equal contribution.)

- ContextInk: Making Free-Form Annotations Reusable in AI-Supported Active Reading. **Yantong Liu***, Xinyu Gai*, Mingyuan Qin, and Mingming Fan. Under review.
- A Survey of Advances in Wake-Up Systems: Toward Context Awareness, Multimodality, and Personalization. Zihong He*, **Yantong Liu***, Yuhui Zheng*, Chun Yu, Chen Liang, Yuanchun Shi, and Hai-Ning Liang. Revised manuscript under review.
- MindWeaver: Scaffolding Critical Thinking Through Structure-Mediated Human–AI Collaboration. Qing Shi, Yilin Zheng, **Yantong Liu**, Luyao Shen, Ran Li, Xuning Zhang, Yi-Lung Kuo, and Mingming Fan. Under review.
- Reflectant: Supporting Reflection on Multi-Source Information through Self-Representation in Multi-Agent Discussions. Zhiyuan Xia, Zhuoyang Jiang, **Yantong Liu**, Chao Liu, Zhuofan Ruan, and Mingming Fan. Under review.

RESEARCH EXPERIENCE

ContextInk: Making Free-Form Annotations Reusable in AI-Supported Active Reading

Supervisor: Prof. Mingming Fan | Feb. 2026 – Present

- Led a formative study using retrospective interviews and analysis of prior artifacts (N = 6) to characterize the uses of visual traces of thinking and derive design principles for reconstructing user intent
- Developed a multimodal reading support system based on CoT multiagent reasoning; organized annotations into editable, source-linked reading threads, making free-form annotations interpretable and reusable by AI
- Conducted a within-subjects study (N = 20) comparing freehand annotation with highlighting; coded interaction logs and applied Wilcoxon signed-rank and paired t-tests with Benjamini–Hochberg correction, finding that AI-supported freehand annotations are more usable in knowledge understanding and idea generation

Context-Aware, Multimodal, and Personalized Wake-Up Systems

Supervisor: Prof. Chen Liang | Oct. 2025 – Present

- Synthesized 116 publications through a co-developed framework of intent-driven interaction-state transitions, examining how engagement decisions adapt across multiparty social settings, intent-driven interactions, and extended standby
- Authored core survey sections on context awareness and personalization, organizing tasks and approaches into a hierarchical taxonomy to examine how contextual cues and user characteristics shape AI engagement
- Trained an edge CNN and cloud Transformer with prediction and consistency objectives to model interaction intent; personalized speech representations using 50k+ XTTS-v2 voice-cloned training samples

Analyzing the Evolution of Public Opinion in Social Media Comments through Hierarchical Visualization

Outstanding Graduation Project, Bachelor Thesis | Supervisor: Prof. Ying Lian | Nov. 2024 – Apr. 2025

- Investigated how topics and emotions evolve through online interactions by reconstructing reply hierarchies for 50k+ Weibo comments; applied LLM labeling with a topic taxonomy informed by pilot interviews and refined manually
- Developed an innovative interactive D3.js visualization combining radial layouts, topic and emotion encodings, and depth filtering to preserve conversational context and trace how discussions shift over time and along reply chains
- Identified more negative sentiment in deeper replies and clusters of discriminatory expressions within the analyzed discussions, illustrating how conversational structure reveals changes in emotional tone and departures from the original topic

PROJECT EXPERIENCE

Understanding Human Collaboration with a Soft Floating Robot during Indoor Inspection Tasks

MPhil Thesis Project | Supervisors: Prof. Mingming Fan and Chen Liang | Sep. 2025 – Present

- Examining coordination in interdependent human–robot tasks, focusing on how mutual awareness of task progress and complementary spatial capabilities support collaboration
- Defined the research scope, led sensing-system development, and co-developed the robot's embodiment and mechanical structure; designed the field deployment protocol and am preparing a user study of these coordination mechanisms
- Coordinate flight control, data analysis, and appearance and interaction design across an interdisciplinary team, aligning prototype development with the study's research questions

Field Research on Elderly Friendly Renovations for Beijing Residents

First Prize, National Market Research Competition (Beijing) | Group Leader | Feb. 2024 – May 2024

- Led a mixed-methods community study examining older adults' perceived effectiveness of and trust in age-friendly renovations; analyzed 696 survey responses and validated the measurement model through exploratory and confirmatory factor analyses
- Applied structural equation modeling and binary logistic regression to examine renovation intentions and adoption, identifying social support as the strongest correlate of intentions in the SEM (standardized $\beta = 0.499$); produced a report offering a communication perspective on municipal eldercare policy

EV Battery Reuse and Mobile Charging in Hong Kong

Group Leader | Feb. 2026 | Hong Kong

- Led a four-member team during a seven-day field program, working with Hong Kong Productivity Council (HKPC) staff to develop a proposal addressing EV battery retirement and charging shortages
- Analyzed government charging-infrastructure data, subsidy policies, and published resident surveys; visualized district-level disparities to identify charging barriers in older housing estates
- Proposed repurposing retired EV batteries into mobile charging units, connecting battery reuse with charging access in residential areas constrained by limited electrical capacity and installation approvals

PROFESSIONAL EXPERIENCE

State Key Laboratory of Media Convergence and Communication

Research Intern | Supervisor: Hao Shen | Jan. 2024 – Jun. 2025

- Developed an automated reporting workflow and visualizations to support analysis for periodic public opinion reports co-produced for the Beijing Municipal Government
- Co-authored the *China Virtual Digital Human Influence Index Report*, analyzing virtual human data and developing key visualizations; contributed to two posters presented at ChinaVis 2024 and delivered fast-forward talks on behalf of the author teams

Caixin Media Company Ltd. VisLab

Data Reporting Intern | Aug. 2023 – Nov. 2023

- Independently published 4 data journalism reports and co-authored 5 on local government debt, disasters, healthcare, education, and culture; one report ranked second on the Digital News board of Caixin
- Contributed to the Caixin Macroeconomic Survey Report through data processing and development of an interactive policy timeline and influence index

CONFERENCE PRESENTATIONS AND SERVICE

Session Moderator, *AI Support for Learning, Recall & Performance*, GBA HCI Pre-CHI 2026
Presenter, fast-forward talk and poster, *ChinaVis 2024*, Hong Kong, Jul. 2024

SELECTED HONORS AND AWARDS

Outstanding Graduation Project, Communication University of China 2025
First Prize, National Market Research Competition (Beijing Division) 2024
Outstanding Student (three consecutive academic years), Communication University of China 2022 – 2025

SKILLS

- **Language**: Mandarin (Native), English (IELTS 7.0)
- **Programming and Engineering**: Python, Java, Basic Embedded Development
- **Qualitative & Quantitative Methods**: Interviews, Thematic Analysis, Survey Design, Statistical Analysis, A/B Testing
- **Design & Visual Analytics**: Figma, Adobe Creative Suite (Ps, Ai); D3.js, ECharts, UI/UX Architecture